



NORTH SOUTH UNIVERSITY

Department of Mathematics and Physics

Assignment # 1, Semester: Spring 2026, Section: 19

Subject: **MAT 120** (Calculus and Analytical Geometry I)

Submission Date & Time: 05/03/2026, Class Time		Credits: 3	Total Marks: 10
Full Name:		ID:	
Instructions: Answer <u>ALL</u> of the following questions. Write your answer in A4 size paper with proper margin. Your answer must be relevant, neat and clean.			

Q1. Sketch the graphs of the following functions:

$$(i) y = \frac{x^2 - 1}{x^4 + 1}, (ii) f(x) = -e^{x-1} + 3, (iii) y = \operatorname{cosec}^{-1} x, (iv) f(x) = -1 + 3^x, (v) y = \log(2 - x^2).$$

From the graph or otherwise, find the domain and range in each case.

Q2. Find the inverse of the functions defined by (i) $f(x) = -\frac{1}{\sqrt{2x-3}}$ and (ii) $f(x) = -1 - \sqrt{1-3x}$; state domain and range of f^{-1} . Also, sketch graph of f and f^{-1} on the same set of axes with reflections of one another about the line $y = x$.

Q3. Compute the limits: i. $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$, ii. $\lim_{y \rightarrow -\infty} \frac{2-y}{\sqrt{7+6y^2}}$, iii. $\lim_{x \rightarrow \frac{1}{2}} \frac{2x-1}{|1-2x|}$, and iv. $\lim_{y \rightarrow -\infty} \frac{\sqrt{y^2+2}}{3-y}$.

Q4. Determine the points, if any, at which the following functions are discontinuous, and sketch their graphs, showing the points of discontinuity.

$$(i) f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases}, (ii) f(x) = \begin{cases} 0, & x < -1 \\ \sqrt{1-x^2}, & -1 < x < 1 \\ x, & x \geq 1 \end{cases}$$

Q5. Justify whether the function $f(x) = -1 - |2-3x|$ is differentiable or not. Also, find a formula for $f'(x)$. Graph the results and explain.

Q6. Find the derivative with respect to x of $f(x) = -1 - \sqrt{1-3x}$ and use it to find the equation of tangent line to $y = -1 - \sqrt{1-3x}$ at $x = -1$. Also, find the limit of $f'(x)$ as $x \rightarrow \frac{1}{3}^-$ and as $x \rightarrow -\infty$, and explain what those limits say about the graph of f .

Q7. Find the values of α and β , if possible, so that the function $f(x) = \begin{cases} 3x^2, & x \leq 1 \\ \alpha x + \beta, & x > 1 \end{cases}$ will be differentiable.

Q8. Find all values of x at which the tangent line to the given curve satisfies the stated property:

(i) $y = \frac{x^2 - 1}{x + 2}$; horizontal, (ii) $y = \frac{x + 3}{x + 2}$; perpendicular to the line $y = x$.

Q9. (i) Apply the Chain rule to find $\frac{d\theta}{d\beta}$ if $\theta = \operatorname{cosec}^2\left(\frac{\pi}{3} - \beta\right) + \mu \cos^2 \pi\beta$. (ii) Compute $F'(\pi/6)$, where $F(x) = [(f \circ g)(x)]^3$, $f(x) = 2\sqrt{x}$, $g(x) = 1 - 3\cos x$. (iii) Find $\frac{d}{dx} [\tan x \sqrt{1 - \sin x}]$.

Q10. The temperature $f(t)$ of the body at time t in appropriate units after being introduced into an environment having constant temperature T_0 is given by the function: $f(t) = T_0 + Ce^{-kt}$, where C and k are constants. A pot of coffee with a temperature of 100°C is set down in a room with a temperature of 25°C . The coffee cools to 55°C after 1 hour. (i) Write an equation to model data. (ii) Find the temperature after half an hour. (iii) How long will it take for the coffee to cool to 42°C .

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